

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

**ENVIRONMENTAL SCIENCE****Paper 1 (Generated practice paper - Set 09)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

**Time allowed: 3 hours****Materials**

For this paper you may use:

- a calculator.

**Instructions**

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

**Information**

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|----------|------|
| 1        |      |
| 2        |      |
| 3        |      |
| 4        |      |
| 5        |      |
| 6        |      |
| 7        |      |
| 8        |      |
| 9        |      |
| 10       |      |
| 11       |      |
| TOTAL    |      |

Answer all questions in the spaces provided.

0 1 . 1

Define the term **persistence** as applied to a pollutant.

[1 mark]

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0 1 . 2

Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

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5

0 2 . 1

Explain the difference between a negative and a positive feedback mechanism.

[2 marks]

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0 2 . 2

Describe **two** negative feedback mechanisms that may reduce global warming.

[4 marks]

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0 2 . 3

Explain how melting permafrost and melting methane hydrate act as positive feedback mechanisms.

[4 marks]

0 2 . 4

Give **three** other positive feedback mechanisms.

[3 marks]

1 .....

2 .....

3 .....

0 2 . 5

Explain what is meant by a climate **tipping point**.

[2 marks]

15

03

Table 1 shows the volume of water in some reservoirs of the hydrological cycle and the annual transfer rates through them.

Table 1

| Reservoir             | Volume / $\times 10^{15}$<br>$\text{dm}^3$ | Annual transfer<br>(inflow = outflow) /<br>$\times 10^{15} \text{ dm}^3 \text{ yr}^{-1}$ | Percentage of<br>global water / % |
|-----------------------|--------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------|
| Oceans                | 1 340 000                                  | 425                                                                                      | 97                                |
| Ice caps and glaciers | 24 000                                     | 3                                                                                        | 1.7                               |
| Groundwater           | 23 000                                     | 12                                                                                       | 1.7                               |
| Lakes and rivers      | 180                                        | 40                                                                                       | 0.013                             |
| Soil moisture         | 17                                         | 80                                                                                       | 0.001                             |
| Atmosphere            | 13                                         | 480                                                                                      | 0.001                             |

03

1

The residence time of water in a reservoir is given by: residence time = volume in reservoir / annual transfer rate. Use Table 1 to calculate the residence time of water in the atmosphere. Give your answer to the nearest whole day.

[3 marks]

Answer

days

03

2

Calculate the residence time of water in the oceans.

[1 mark]

Answer  years

**0 3 . 3** Explain why the atmosphere is an important reservoir even though it holds only 0.001% of global water. **[2 marks]**

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**0 3 . 4** Explain why human changes to a small reservoir such as soil moisture are more obvious than changes to the oceans. **[2 marks]**

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**0 3 . 5** Explain why the hydrological cycle is described as being in dynamic equilibrium. **[2 marks]**

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**10**

0 4 . 1

Describe **three** factors that have restricted the growth of nuclear power.

[3 marks]

1 .....

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2 .....

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3 .....

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0 4 . 2

Explain how a plutonium 'breeder' reactor can produce more fissile fuel than it uses.

[3 marks]

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0 4 . 3

Give **two** advantages and **two** disadvantages of thorium reactors compared with uranium reactors.

[4 marks]

**Advantages** .....

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**Disadvantages** .....

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.....

10

05 . 1

Which of the following best describes a **representative** sample? Tick one box.**[1 mark]**

A A sample taken from the most convenient location

☐

B A sub-sample that accurately reflects the complete data set

☐

C The largest sample that can be collected

☐

D A sample that supports the hypothesis

☐

05 . 2

Describe how random sampling sites would be selected in an irregularly shaped meadow.

**[2 marks]**

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05 . 3

Explain why deliberately choosing sampling sites introduces bias.

**[2 marks]**

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5

06 . 1

Explain how each of the following environmental factors affects the rate of degradation of pollutants. Give a named example of a pollutant for each factor.

**[6 marks]**

**Temperature** .....

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**Light** .....

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**Oxygen** .....

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**06 . 2**

Explain how the presence of adsorbent materials such as clay can reduce the severity of pollution by heavy metals or radioactive caesium-137.

**[2 marks]**

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**06 . 3**

Explain why a fall in pH increases the harm caused by many heavy metals.

**[2 marks]**

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**10**

07 . 1

Describe **four** factors that should be considered when choosing a site for a dam and reservoir.

[4 marks]

1 .....

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2 .....

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3 .....

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4 .....

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07 . 2

Give **one** advantage and **two** disadvantages of an estuary barrage compared with a reservoir on land.

[3 marks]

Advantage .....

.....

Disadvantage 1 .....

.....

Disadvantage 2 .....

.....

07 . 3

Explain why desalination of seawater is only used in some countries.

[2 marks]

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07 . 4

Explain how artificial recharge can make aquifer supplies sustainable.

[1 mark]

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**0 8** Table 1 shows typical noise levels of some sounds.

**Table 1**

| Sound                                   | Noise level / dB |
|-----------------------------------------|------------------|
| Threshold of human hearing (at 1000 Hz) | 0                |
| Leaves rustling                         | 20               |
| Library                                 | 40               |
| Normal conversation                     | 60               |
| Busy road                               | 80               |
| Chain saw                               | 110              |
| Jet engine at 30 m                      | 140              |

**0 8** . **1** Explain why a logarithmic scale is used to measure sound.

**[2 marks]**

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**0 8** . **2** Use Table 1 to calculate how many times greater the sound intensity of a chain saw is than that of a busy road.

**[2 marks]**

**Answer**  times

**0 8** . **3** A machine produces 85 dB. A second identical machine is switched on next to it. State the new noise level and explain your answer.

**[2 marks]**

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08 . 4

Explain what is meant by the statement that 0 dB does not mean there is no sound.

[1 mark]

08 . 5

State the frequency range of hearing for a typical young person and explain why the dB(A) scale is used for human hearing.

[3 marks]

10

09

Table 1 shows information about two copper deposits, X and Y, being assessed for mining. The current cut-off ore grade for copper is 0.4%.

Table 1

| Feature                          | Deposit X         | Deposit Y              |
|----------------------------------|-------------------|------------------------|
| Ore grade / % copper             | 0.9               | 0.35                   |
| Depth of deposit / m             | 350               | 40                     |
| Overburden                       | Hard igneous rock | Loose glacial sediment |
| Distance to nearest railway / km | 120               | 8                      |
| Annual rainfall / mm             | 2 100             | 500                    |
| Chemical form                    | Copper sulfide    | Copper sulfide         |

09 . 1

Explain what is meant by the **cut-off ore grade** and state which deposit is currently included in the copper reserves.

[2 marks]

|   |   |   |   |
|---|---|---|---|
| 0 | 9 | . | 2 |
|---|---|---|---|

Use Table 1 to discuss the factors that affect the viability of mining each deposit.

**[6 marks]**

|   |   |   |   |
|---|---|---|---|
| 0 | 9 | . | 3 |
|---|---|---|---|

State **two** changes that would decrease the cut-off ore grade.

**[2 marks]**

**10**

Explain how the properties of smoke and of the gases involved in acid rain lead to their environmental effects, and evaluate the effectiveness of the methods used to control them.

ESP/7447/1/SET09

Extra space

**10** . **2** State **one** effect of tropospheric ozone on plants.

**[1 mark]**

**10**

**Write an essay on one of the following topics.**

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

**[25 marks]**

**OR**

**1 1 . 2**

Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

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A large rectangular box containing 30 horizontal dotted lines for writing.

**END OF QUESTIONS**